

1

Here are the equations of two straight lines.

$$y = \frac{1}{2}x - 6 \qquad 6y = 3x + 7$$

Oscar says that these lines are parallel.

Is Oscar correct?

You must give a reason for your answer.

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(Total for Question 1 is 2 marks)

- 2 The equation of the line L_1 is $y = 3x - 2$
The equation of the line L_2 is $3y - 9x + 5 = 0$

Show that these two lines are parallel.

(Total for Question 2 is 2 marks)

- 3** The equation of the line L_1 is $y = 2x + 3$
The equation of the line L_2 is $5y - 10x + 4 = 0$
Show that these two lines are parallel.

(Total for Question 3 is 2 marks)

4 A and B are points on a centimetre grid.

A is the point with coordinates $(-7, 6)$

B is the point with coordinates $(8, -5)$

Work out the length of AB .

Give your answer correct to 1 decimal place.

..... cm

(Total for Question 4 is 2 marks)

- 5** A is the point with coordinates $(5, 9)$
 B is the point with coordinates $(d, 15)$

The gradient of the line AB is 3

Work out the value of d .

.....
(Total for Question 5 is 3 marks)

- 6** A is the point with coordinates $(7, 13)$
 B is the point with coordinates $(-3, 21)$
 C is the point with coordinates $(15, 23)$

M is the midpoint of AB .

N is the midpoint of BC .

Work out the distance between M and N .

Give your answer correct to 1 decimal place.

(Total for Question 6 is 3 marks)

- 7 The straight line L_1 has equation $y = 3x - 4$
The straight line L_2 is perpendicular to L_1 and passes through the point $(9, 5)$
Find an equation of line L_2

(Total for Question 7 is 3 marks)

- 8** The straight line **L** has the equation $3y = 4x + 7$
The point *A* has coordinates $(3, -5)$

Find an equation of the straight line that is perpendicular to **L** and passes through *A*.

(Total for Question 8 is 3 marks)

9 The straight line **L** has equation $2y = 3x - 7$

Find an equation of the straight line perpendicular to **L** that passes through $(6, -5)$

(Total for Question 9 is 3 marks)

- 10** The equation of line L_1 is $y = 2x - 5$
The equation of line L_2 is $6y + kx - 12 = 0$

L_1 is perpendicular to L_2

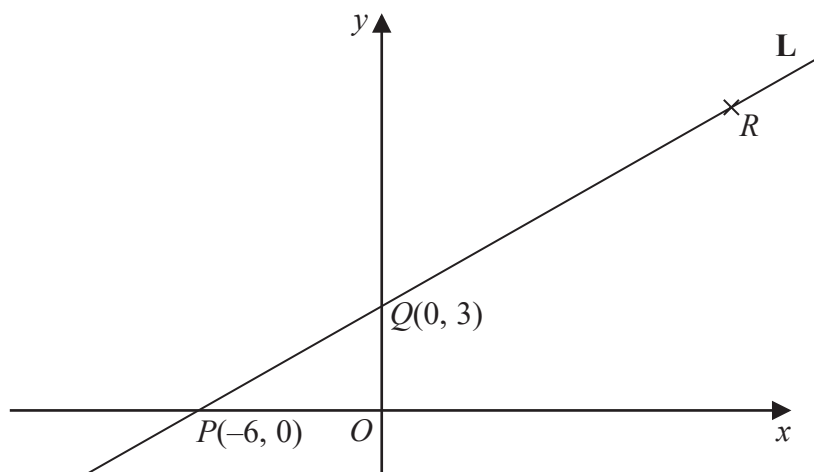
Find the value of k .

You must show all your working.

$k = \dots\dots\dots$

(Total for Question 10 is 3 marks)

11 Here is a sketch of the line **L**.



The points $P(-6, 0)$ and $Q(0, 3)$ are points on the line **L**.

The point R is such that PQR is a straight line and $PQ:QR = 2:3$

(a) Find the coordinates of R .

(.....,)
(2)

(b) Find an equation of the line that is perpendicular to **L** and passes through Q .

.....
(3)

(Total for Question 11 is 5 marks)

12 The points L , M and N are such that LMN is a straight line.

The coordinates of L are $(-3, 1)$

The coordinates of M are $(4, 9)$

Given that $LM : MN = 2 : 3$,

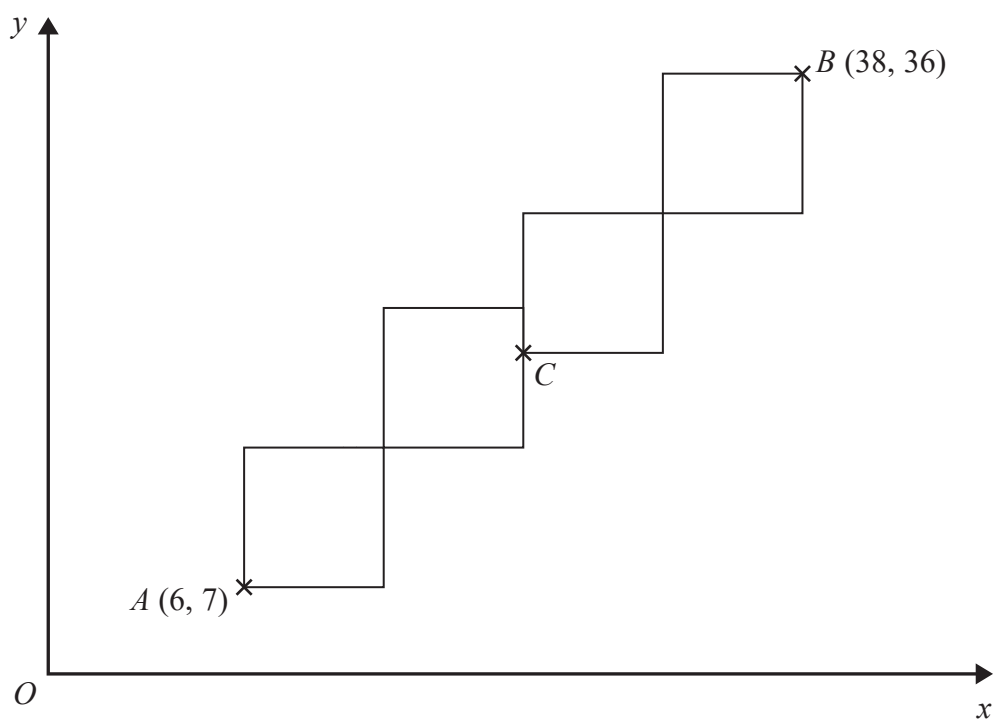
find the coordinates of N .

(..... ,)

(Total for Question 12 is 4 marks)

13 A pattern is made from four identical squares.

The sides of the squares are parallel to the axes.



Point *A* has coordinates (6, 7)

Point *B* has coordinates (38, 36)

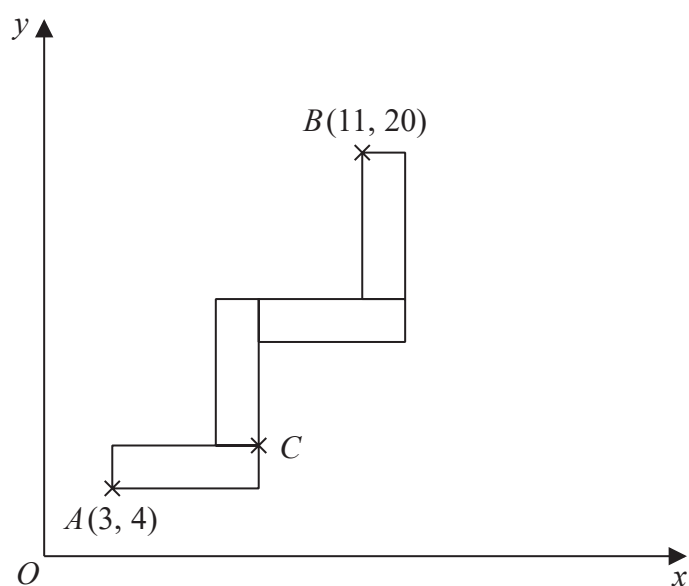
Point *C* is marked on the diagram.

Work out the coordinates of *C*.

(..... ,)

(Total for Question 13 is 5 marks)

- 14 A pattern is made from four identical rectangles.
The sides of the rectangles are parallel to the axes.



Point A has coordinates $(3, 4)$
 Point B has coordinates $(11, 20)$
 Point C is marked on the diagram.

Work out the coordinates of C .
 You must show all your working.

(..... ,)

(Total for Question 14 is 5 marks)